

Andrew Stout

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RESEARCH INTERESTS

Algebraic Geometry, Commutative Algebra, Computer Algebra, Combinatorics.

EXPERIENCE

- **Borough of Manhattan Community College, The City University of New York (CUNY)** [🌐] *Aug 2023 - Current*
Associate Professor, Tenured NYC, NY
- **Borough of Manhattan Community College, CUNY** [🌐] *Aug 2016 - Aug 2023*
Assistant Professor, Tenure-Track NYC, NY
- **Bronx Community College, CUNY** [🌐] *Aug 2014 - Aug 2016*
Substitute Assistant Professor NYC, NY
- **Université Pierre-et-Marie-Curie, Sorbonne Université** [🌐] *Aug 2012 - Aug 2013*
Chateaubriand Fellowship Paris, Fr
- **The Graduate School and University Center, CUNY** [🌐] *Aug 2008 - Dec. 2011*
Enhanced Chancellor's Fellowship NYC, NY

EDUCATION

- **The Graduate School and University Center, CUNY** [🌐] *May 2014*
Ph.D., Mathematics NYC, NY
- **The Graduate School and University Center, CUNY** [🌐] *May 2011*
M.A., Mathematics NYC, NY
- **North Carolina State University** [🌐] *May 2008*
B.Sc., Mathematics Raleigh, NC

PUBLICATIONS & ABSTRACTS

A=ABSTRACT, C=CODE, J=JOURNAL ARTICLE, S=IN SUBMISSION, T=THESIS

- [J.6] Stout, A. (2025). **Jet schemes of local complete intersection morphisms**, 25pp. J. Lond. Math. Soc., Accepted, In-press.
- [S.5] Stout, A. (2025). **arc.lib: A Singular library for jet schemes with fast partial reduction**, SoftwareX, Volume 32, 102420.
- [J.4] Stout, A. (2024). **Jet schemes over auto-arc spaces and deformations of locally complete intersection varieties**, J. Algebra, Volume 659, 361-394
- [J.3] Stout, A. (2019). **The auto Igusa-zeta function of a plane curve singularity is rational**. Proc. Amer. Math. Soc., 147, no. 5, 1825–1838.
- [J.2] Stout, A. (2019). **Formal deformations of algebraic spaces and generalizations of the motivic Igusa-zeta function**. Algebraic curves and their applications, 137–147, Contemp. Math., 724, Amer. Math. Soc.,
- [J.1] Stout, A. (2017). **On the auto Igusa-zeta function of an algebraic curve**. J. Symbolic Comput. 79, part 1, 156–185.
- [C.2] Stout, A. (2025), **arc.lib: A Singular library for generalized jet schemes computations**, Version 1.0.0, June 2025. Additional repository at <https://github.com/infinity-groupoids/arc.lib>
- [C.1] Stout, A. (2014), **A sage script entitled "NFT.PY" which computes generalized jet schemes**, Version 1.0.0, May 2014. Additional repository at <https://github.com/infinity-groupoids/NFT.py>
- [T.1] Stout, A. (2014). **Motivic Integration over Nilpotent Structures**. Thesis (Ph.D.)—City University of New York. 179 pp. ISBN: 978-1321-30704-7, ProQuest LLC
- [A.3] Stout, A., **The jet operator: from local to global deformations**. AMS Special Session on Integer Partitions, Arc Spaces and Vertex Operators, Joint Mathematics Meeting, San Francisco, CA, United States. Jan. 2024
- [A.2] Stout, A. with Iarmolenko, A., Legaspi, M., Vaiciulis, P., **Using the Gini Coefficient to rank AI systems: a case study using Rubik's cube AI solvers**. AMS - PME Undergraduate Student Poster Session, American Mathematical Society, Joint Mathematics Meeting, San Francisco, CA, United States. Jan. 2024
- [A.1] Stout, A., **The auto Igusa-zeta function of a plane curve singularity is rational**. Special Session on Algebraic Curves and their Applications, American Mathematical Society, Fall Southeastern Sectional Meeting, University of Central Florida Orlando, FL, United States. Sept. 2017.

TEACHING EXPERIENCE

- **BMCC - CUNY:** MAT 415: Linear Algebra for Data Science, MAT 320: Abstract Algebra, MAT 315: Linear Algebra, MAT 301: Analytic Geometry & Calculus I, MAT 209: Statistics, Mat 206: Pre-Calculus, MAT 150: Introduction to Statistics, MAT 56: Intermediate Algebra and Trigonometry
- **BCC - CUNY:** Mat 23: Introduction to Statistics, Mat 05: Remedial Algebra, Mat 01: Remedial Arithmetic
- **Stevens Institute of Technology:** MA 530: Applied Mathematics for Engineers & Scientists II
- **NYU:** Introduction to Statistics
- **Hunter College - CUNY:** Calculus 1, Precalculus
- **City Tech - CUNY:** Calculus 1, Precalculus
- **CCNY - CUNY:** Precalculus, College Algebra

GRANTS & AWARDS

- **Award: Faculty Development Grant** Fall 2025 - Spring 2026
Borough of Manhattan Community College, CUNY 
- **PSC- CUNY Award #67566-00 Cycle 55 (Trad B), PI** Fall 2024
Mathematics Research, PSC-CUNY 
- **Creating Data Science Pathways for STEM Student Success, Co-PI** Spring 2024 - Present
Program Advisor, NSF Grant 
- **Creating Data Science Pathways for STEM Student Success Summer Grant, PI** Summer 2023
Guiding Student Research, NSF Grant 
- **PSC- CUNY Award #66024-00 Cycle 54 (Trad A), PI** Summer 2023
Mathematics Research, PSC-CUNY 
- **PSC- CUNY Award #60784-00 Cycle 48 (Trad B), PI** Fall 2017
Mathematics Research, PSC-CUNY 

SELECTED DEPARTMENT, COLLEGE & UNIVERSITY SERVICE

- **Member of Focus Group** Jan 2025 - May 2025
Shape NYCPS + CUNY Data Science Pathways, University Service 
- **Program Advisor** Dec 2023 - Dec 2024
Data Science Program, Department Service 
- **Chair** Aug 2023 - Dec 2024
Data Science Committee, Department Service 
- **Member** Aug 2023 - Dec 2024
Data Science Team, Department Service 
- **Member** Aug 2022 - Present
AcMo 2.0 Committee, College Service 
- **Event Organizer** Aug 2019 - Present
Event planning for talks and research symposiums, Department & College Service 

STUDENT MENTORSHIP

- **Guiding Student Research in Math & Data Science** Aug 2017 - Present
Organization/Institution Name 
 - Mentored one student for honors contract and research (Spring 2024).
 - One student in my research team presented our work at JMM 2024 in the AMS-PME Poster Session.
 - Guiding students on where to apply for research funding and other academic programs.
 - During Summer 2023, I mentored three students in research in Artificial Intelligence and Deep Learning which resulted in Poster Presentation at the Data Science Symposium.
 - During Fall 2023 and Spring 2024, I mentored one of the students for BMCC Foundation Fellowship to continue to conduct research in Data Science and present at BARs Conferences.
 - Mentored one student for honors contract and research (AY 2017). This student continued on to a four-year school, and I continued to mentor him in research for future Poster Presentations in Mathematics.
- **Mentor** Aug 2023 - Dec 2024
NOYCE Student Fellowships 
 - Met regularly with an advanced student to discuss teaching mathematics (Calculus 1) and how to mentor/guide students learning new content.
 - Scheduled specific tutoring times for students who are struggling in the course to meet with my mentee.
 - Guided my student through the process of creating a short lesson plan for Calculus I in order to prepare for a teaching session in my class.
 - Provided instructional feedback after their brief teaching session to help improve their instructional abilities and confidence.

• Faculty Student Advisor

Aug 2016 - Present

Various Activities



- Advised students (from Fall 2023 to Present) in Data Science in terms of which class to take.
- Faculty Chaperone for 1 day student club trip to Brooklyn Botanical Gardens during Fall 2019.
- Faculty Chaperone for 3 day trip student club trip to University Of Pennsylvania during Spring 2019.
- Faculty club advisor: Chess Club during Spring 2019.
- Faculty club advisor: Investment Club during Spring 2019.
- Advised students of all majors in terms of which classes to take from Fall 2016 to Fall 2018.

RECENT TALKS

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| • Computations of generalized jet schemes in SINGULAR: CAAG Seminar, Graduate Center, CUNY | Oct. 2025 |
| • Jets of lci morphisms: Algebraic Geometry Seminar, Duke University | Oct. 2025 |
| • Computations of generalized jet schemes in SINGULAR: ANT-COG Seminar, UNCG | Sep. 2025 |
| • Intro to Jet Schemes: Math Lounge Colloquium, BMCC, CUNY | May 2025 |
| • Jets of lci morphisms: Algebra Seminar, Rutgers University | April 2025 |
| • Intro to Jet Schemes: Mathematical Research Series, Kean University | March 2025 |
| • Jets of lci morphisms: CAAG Seminar, Graduate Center, CUNY | Feb. 2025 |
| • Jet Spaces: from local to global deformations: MAA Metro NYC Conference, City Tech, CUNY | April. 2024 |
| • Using the Gini coefficient to rank AI systems: Joint Mathematics Meeting, San Francisco | Jan. 2024 |
| • Jet Spaces: from local to global deformations: CAAG Seminar, Graduate Center, CUNY | Oct. 2023 |
| • Using the Gini coefficient to rank AI systems: Annual Data Science Symposium, BMCC, CUNY | Sep. 2023 |

PROFESSIONAL MEMBERSHIPS

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- American Mathematical Society
 - Mathematical Association of America

SELECTED SERVICE TO MATHEMATICAL COMMUNITY

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- Reviewer for Mathematical Reviews, American Mathematical Society: mathscinet.ams.org (2017 - Present)
 - Referee for Journal de l'École Polytechnique - Mathématiques
 - Postdoctoral Grant Evaluator for European Science Foundation - Research Foundation Flanders' (FWO)

ADDITIONAL INFORMATION

Programming Languages: Python, R, Singular, Sage, Macaulay, Magma, Latex, Java, HTML